

This poster in 10 seconds

SSIM from Frame MSE using one source-dependent factor

SSIM $\approx 1 - k \times \text{MSE}_G$

what we want SSIM starts near 1 computed once per source measured once per encode

k collapses all the expensive local analysis into a single scalar you compute once.

Motivation: fast SSIM approximation

Encoding workflows often compare many candidate encodes of the same source. Full SSIM requires local window statistics for each candidate output [1].

We instead ask: can the encoded frame contribute only global MSE, while the expensive local analysis is computed once from the source?

Method: use frame MSE, reuse source statistics

Prior work [2] showed that local SSIM for DCT-compressed content can be approximated from local MSE and local source variance:

$$\text{SSIM}(x_j, y_j) \approx 1 - \frac{\text{MSE}_\ell}{2\sigma_x^2 + C_2}.$$

In this poster, the encoded frame contributes only its frame MSE, MSE_G . We estimate local MSE by redistributing MSE_G according to source activity:

$$\text{MSE}_\ell \approx \text{MSE}_G \times \frac{(\sigma_x^2 + \epsilon)^\beta}{\mathbb{E}[(\sigma_x^2 + \epsilon)^\beta]}.$$

Since the best β is close to 0.5, we use the parameter-free standard-deviation model:

$$\text{MSE}_\ell \approx \text{MSE}_G \times \frac{\sigma_x + \epsilon}{\mathbb{E}[\sigma_x + \epsilon]}.$$

Deriving the source-only factor k

Substituting the standard-deviation model into the local SSIM approximation and averaging over all windows gives:

$$\text{MSSIM}(X, Y) \approx 1 - \text{MSE}_G \cdot \underbrace{\frac{1}{M} \sum_j \frac{\sigma_x + \epsilon}{\mathbb{E}[\sigma_x + \epsilon](2\sigma_x^2 + C_2)}}_k.$$

Interpretation: k is source-only. Once k is computed for a source image or frame, every candidate encode only needs a global MSE value to estimate SSIM.

Experimental setup

- Datasets:** Kodak Lossless True Color Image Suite (24 images) and Xiph.org Foundation Subset1 still-image test set (50 images).
- Compression:** JPEG as a representative DCT-based codec, quality values from 35 to 95 in steps of 5.
- Reference SSIM:** scikit-image implementation with default 7×7 local windows.
- Compared methods:** global-MSE baseline, local-MSE approximation, sublinear variance weighting, and standard-deviation weighting.
- Reproducibility:** the version used for testing is available as open-source software [3].

Results summary

< 1% error
over the practical quality
range

Monotonic
JPEG quality and
estimated SSIM stay
ordered

low-quality limit
artifacts become less
source-driven

Over the evaluated practical quality range, both proposed approximations remain close to SSIM and preserve the monotonic relationship between JPEG quality and SSIM. At very low quality, distortion becomes increasingly artifact-dependent and less well described by source-only local statistics.

Takeaways

- A single scalar k can capture the source-dependent component needed to estimate SSIM from frame MSE, MSE_G .
- The optimized exponent is close to $\beta \approx 0.5$, supporting a standard-deviation weighting model.
- The approximation preserves JPEG-quality ordering over the practical range, but errors increase at very low quality.

References

- [1] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: from error visibility to structural similarity," *IEEE TIP*, 13(4):600–612, 2004.
- [2] M. G. Martini, "Measuring objective image and video quality: On the relationship between SSIM and PSNR for DCT-based compressed images," *IEEE TIM*, 74:1–13, 2025.
- [3] L. Trudeau and M. G. Martini, "ApproxSSIMate: Estimating SSIM from PSNR/MSE and reference-derived statistics," 2026. Open-source software, version 0.1.0, BSD-2-Clause license.  github.com/luc Trudeau/approxssimate.

Qualitative behavior of the approximation

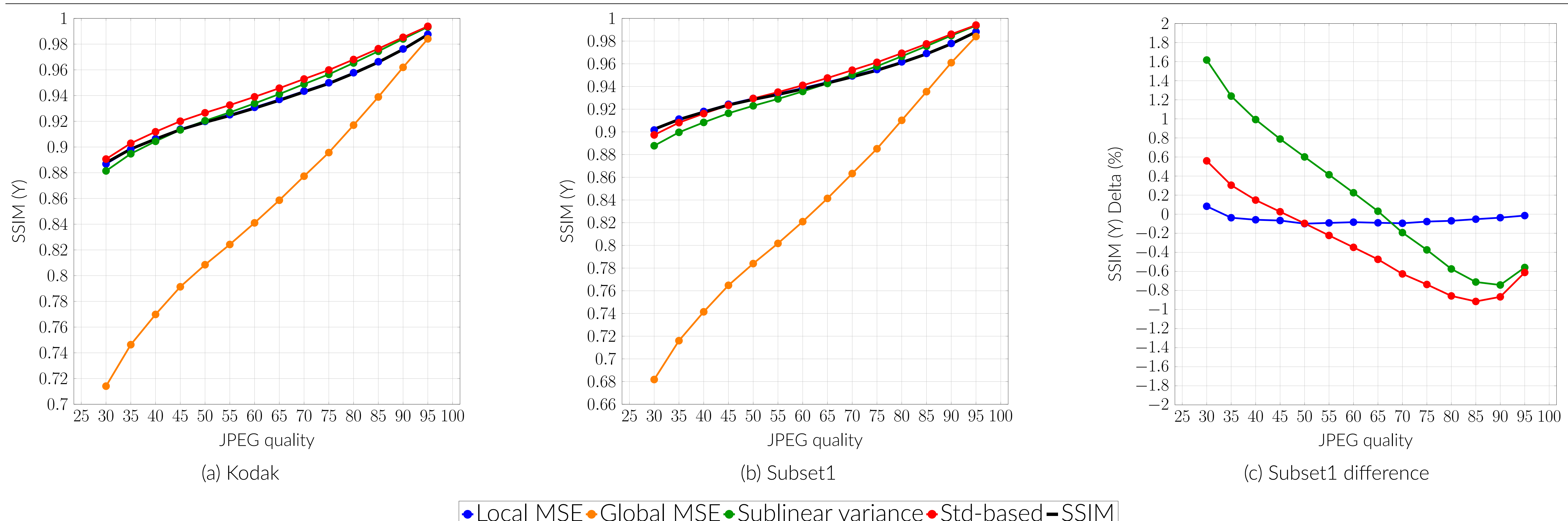


Figure: SSIM and approximations versus JPEG quality on Kodak and Subset1. The proposed source-statistics approximations (green and red) closely follows SSIM, while the global-MSE (orange) baseline deviates more strongly.